El Hajj Malick Diagne

Robotics Engineering and Physics Student

Columbus, OH ehmdiagne0530@email.com ehmd28.github.io EHMD28

Objective

Aspiring robotics software engineer and researcher with a interest in embedded systems, low-level programming, and computational physics. Enjoys learning skills using a project-based approach and understanding how technology works internally.

Education

Metro Early College High School, STEM-focused High School — Columbus, OH
- GPA: 4.23 (weighted) — Aug 2023 – present
- Completed the DESIGN (engineering) pathway and a capstone project.

The Ohio State University, Early College Experience — Columbus, OH
- GPA: 4.00 — Jan 2025 – present

Experience

The Metrobots, Programming Manager — Columbus, OH
- Assisted with programming multiple robots which competed in First Robotics Competition (FRC) competitions. — Aug 2024 – present, 2 years 3 months
- Led two simultaneous outreach projects to teach middle and high school students about digital electronics.
- Overhauled the team website to use a modern web development framework (SvelteKit).

Metro Early College High School, High School Mathematics Tutor — Columbus, OH
- Tutored students in trigonometry, precalculus, and calculus work to assist students in grasping the course material. — Spring 2025 - 2026
- Answered questions for students when the teacher was unavailable, which included re-teaching concepts from previous courses.

Projects

Astrophysics Research Capstone — Spring 2026
- Developed an interactive web application using Python to model the structure of neutron-star matter under the mentorship of a tenured Ohio State University physicist.
- Utilized industry-standard data science and computational modeling libraries (Numpy, SciPy, Matplotlib, and Pandas).

Skills

Programming Languages: C, Rust, Python, TypeScript, Kotlin, Java

Platforms & Frameworks: Arduino, ESP32, Raspberry Pi, SvelteKit, Android

Software: Onshape (3D Modeling), KiCad (Printed Circuit Board Design), Linux (Command-Line Interface)

Areas of Interest

Robotics control systems, swarm robot algorithms, kinematics simulations, physics and mathematics visualizations.